How do RNNs encode mixtures? One network yields distinct geometries

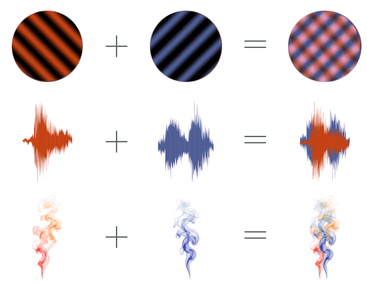
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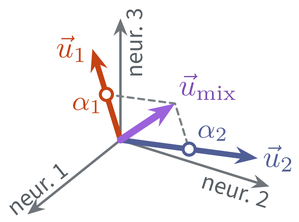
SCAN FOR supporting details, references, contacts
esther-poniatowski.github.io/royaumont2026

1 PROBLEM
Natural inputs often mix simpler components

Weighted sum of two components: $\vec{u}_{\text{mix}} = \alpha_1 \vec{u}_1 + \alpha_2 \vec{u}_2$

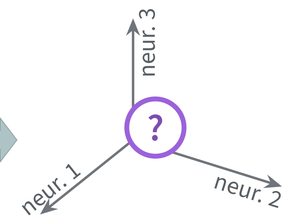
Vision overlapping gratings
Audition mixed voices
Olfaction odour blends





Input vector

recurrent network

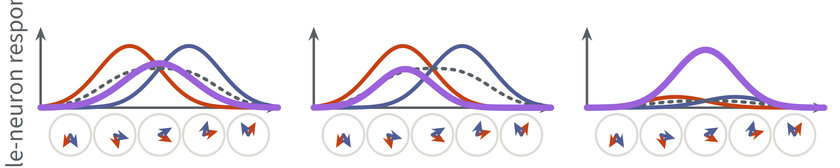


Output vector

How does a whole network encode a mixture of inputs?

2 OBSERVATION
Single neurons respond heterogeneously

Average of both component responses	Response closer to one component	Mixture-specific response
Xiao & Huang, 2015; Zoccolan, 2005	Xiao & Huang, 2015; Reynolds, 1999	Sadagopan & Wang, 2009; Juavinett & Callaway, 2015



Stimulus (circles): a pair of orientations, alone or superimposed

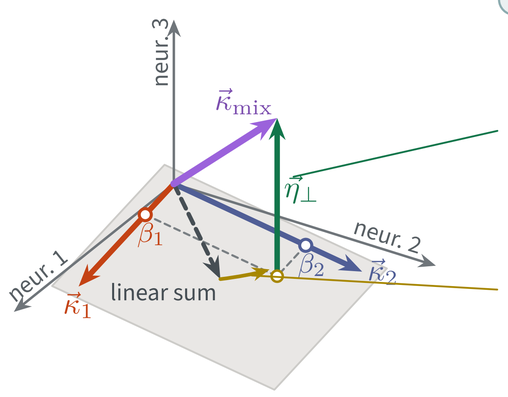
component 1 · component 2 · mixture · linear sum

3 DESCRIPTION FRAMEWORK
The mixture geometry is compared to the components'

Reference: the outputs to each component alone, $\vec{\kappa}_1$ and $\vec{\kappa}_2$, and their linear sum $\alpha_1\vec{\kappa}_1 + \alpha_2\vec{\kappa}_2$.

$$\vec{\kappa}_{\text{mix}} = \underbrace{\beta_1\vec{\kappa}_1 + \beta_2\vec{\kappa}_2}_{\text{in the components plane}} + \underbrace{\beta_\perp\vec{\eta}_\perp}_{\text{orthogonal to the plane}}$$

TWO POTENTIAL DEVIATIONS from the linear sum



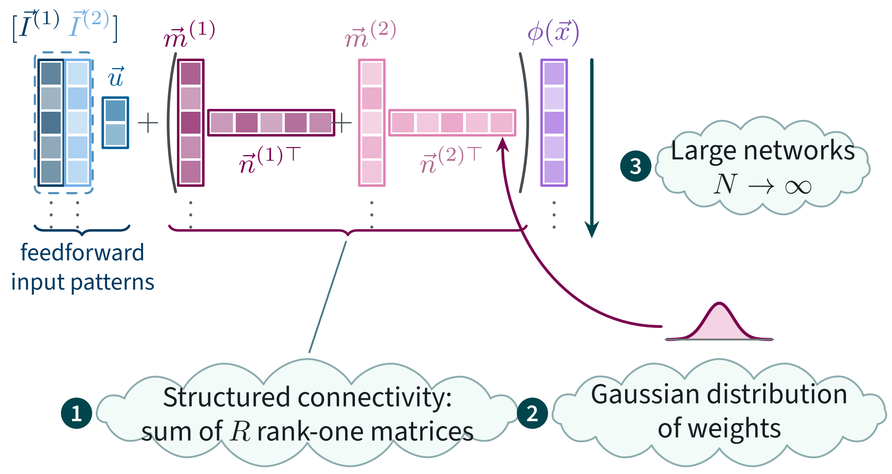
EMERGENCE out of the components plane $\beta_\perp \neq 0$

BIAS relative to the components weights $\beta_1{:}\beta_2 \neq \alpha_1{:}\alpha_2$

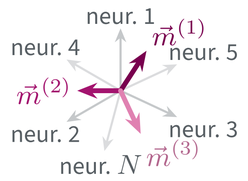
4 MODELING APPROACH
A reduced model links network structure to geometry

A few biologically plausible assumptions reduce a standard rate model to a set of collective statistics (Mastrogiuseppe & Ostojic, 2018; Beiran et al., 2021; Dubreuil et al., 2022).

$$\tau\,\dot{\vec{x}} = -\vec{x} + \left[\vec{I}^{(1)}\ \vec{I}^{(2)}\right]\vec{u} + \frac{1}{N}\sum_{r=1}^{R}\vec{m}^{(r)}\vec{n}^{(r)\top}\phi(\vec{x})$$

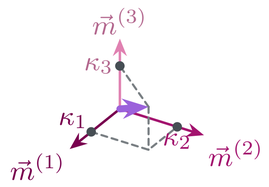


High-dimensional neural space



mean-field reduction

Reduced coordinates capture output geometry

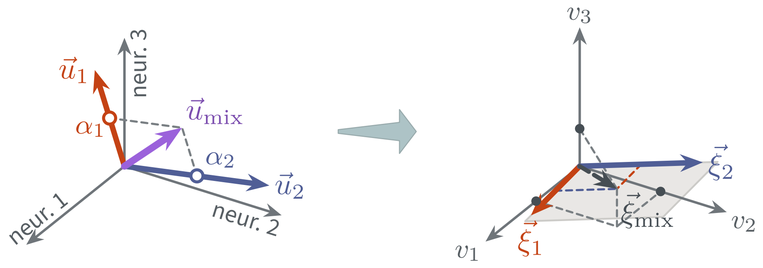


5 MECHANISM
Two network operations determine the geometry

Both operations are explicit along the principal directions of activity fixed by connectivity.

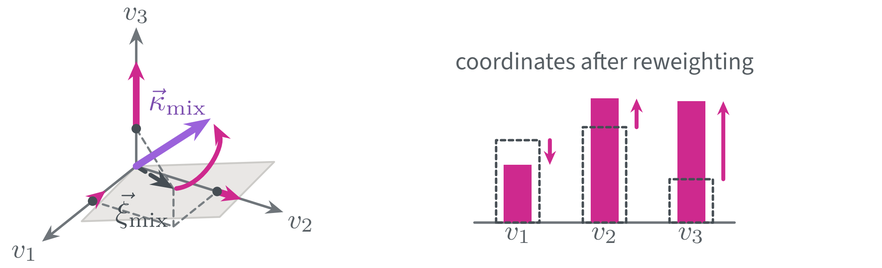
FEEDFORWARD PROJECTION

maps inputs onto population patterns
↓
linear combination of the components
$\vec{\xi}_{\text{mix}} = \alpha_1\vec{\xi}_1 + \alpha_2\vec{\xi}_2$



RECURRENT REWEIGHTING

modulates the contribution of each elicited pattern
↓
rotation relative to the linear sum
$\vec{\kappa}_{\text{mix}} = A_1\,\xi_{\text{mix},1}\,\vec{v}_1 + A_2\,\xi_{\text{mix},2}\,\vec{v}_2 + A_3\,\xi_{\text{mix},3}\,\vec{v}_3$



Bars: coordinates along the principal directions v_k
component 1 · component 2 · projected mixture · after reweighting

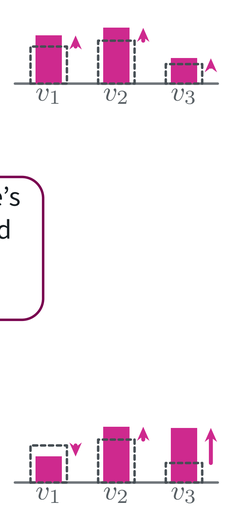
FEEDFORWARD PROJECTION

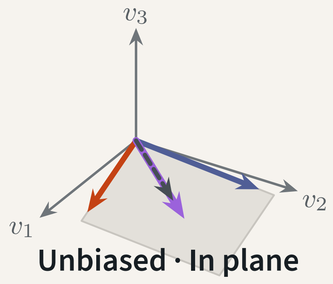
How many eigendir. do the components activate jointly?



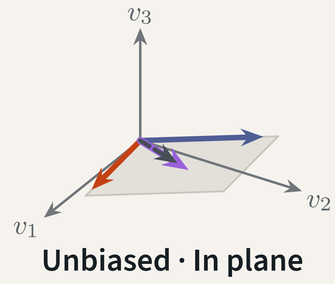
RECURRENT REWEIGHTING

How is the mixture's projection rescaled across the active eigendirections?

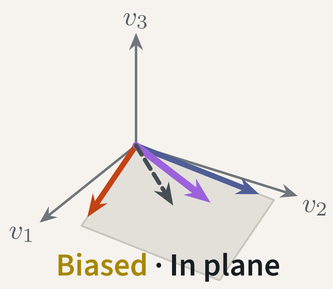




Unbiased · In plane
output aligned with the linear sum

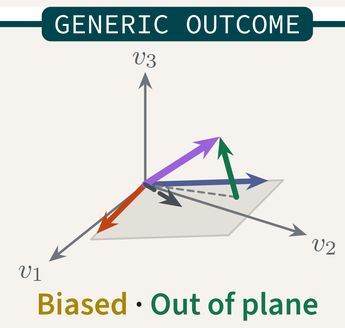


Unbiased · In plane
output aligned with the linear sum



Biased · In plane
output rotates within the plane

GENERIC OUTCOME



Biased · Out of plane
output rotates out of the plane

6 INTERPRETATIONS
Each geometric property shapes what a readout can infer

- A **biased** combination can implement attentional selection by **gain control**.
- An **emergent** pattern is **linearly separable** from its components, readable as a distinct identity.

7 PREDICTIONS
The model predicts testable signatures

- **Emergence requires dissimilar components**, and grows as they separate.
- **Bias decreases** with stimulus intensity.

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